

OS Problem Set 2 Screenshots – Ramis Kamlani 29071

Using gdb

```
0x0000000000000100 in ?? ()
(gdb) b syscall
Breakpoint 1 at 0x80001cde: file kernel/syscall.c, line 133.
(gdb) c
Continuing.
[Switching to Thread 1.3]

Thread 3 hit Breakpoint 1, syscall () at kernel/syscall.c:133
133 {
(gdb)
```

```
ramisk@Ramis-Laptop: ~/xv6
kernel/syscall.c
125 [SYS_unlink] sys_unlink,
126 [SYS_link] sys_link,
127 [SYS_mkdir] sys_mkdir,
128 [SYS_close] sys_close,
129 };
130
131 void
132 syscall(void)
133 {
134     int num;
135     struct proc *p = myproc();
136
137     num = p->trapframe->a7;
138     if(num > 0 && num < NELEM(syscalls) && syscalls[num]
139         // Use num to lookup the system call function fo
140         // and store its return value in p->trapframe->a
141         p->trapframe->a0 = syscalls[num]();
142     } else {
remote Thread 1.3 (src) In: syscall L133 PC: 0x80001cde
(gdb) backtrace
#0  syscall () at kernel/syscall.c:133
#1  0x0000000080001a9c in usertrap () at kernel/trap.c:68
#2  0x00000003ffffff09c in ?? ()
(gdb)
```

```
0x69, 0x6e, 0x69, 0x74, 0x0 <repeats 12 times>}}
(gdb) p /x p->trapframe->a7
$2 = 0xf
(gdb) p /x $sstatus
$3 = 0x200000022
(gdb)
```

xv6 kernel is booting

```
hart 2 starting
hart 1 starting
scause=0xd sepc=0x80001cee stval=0x0
panic: kerneltrap
QEMU: Terminated
```

```
ramisk@Ramis-Laptop: ~/xv6
GNU nano 7.2 kernel.asm
void
syscall(void)
{
    80001cde: 1101          addi    sp,sp,-32
    80001ce0: ec06          sd      ra,24(sp)
    80001ce2: e822          sd      s0,16(sp)
    80001ce4: e426          sd      s1,8(sp)
    80001ce6: 1000          addi    s0,sp,32

    int num;
    struct proc *p = myproc();
    80001ce8: 892ff0ef      jal     80000d7a <myproc>
    80001cec: 84aa          mv      s1,a0

    num = * (int *) 0;
    80001cee: 00002683      lw      a3,0(zero) # 0 <_entry-0x80000000>
    if(num > 0 && num < NELEM(syscalls) && syscalls[num]) {
    80001cf2: fff6871b      addiw   a4,a3,-1 # fff <_entry-0x7ffff001>

```

```

ramisk@Ramis-Laptop: ~/xv6
o-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -
fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I.
-fno-stack-protector -fno-pie -no-pie -c -o user/secret.o use
r/secret.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o us
er/_secret user/secret.o user/ulib.o user/usys.o user/printf.o u
ser/umalloc.o
riscv64-linux-gnu-objdump -S user/_secret > user/secret.asm
riscv64-linux-gnu-objdump -t user/_secret | sed '1,/SYMBOL TABLE
/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_
forktest user/_grep user/_init user/_kill user/_ln user/_ls user
/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_g
rind user/_wc user/_zombie user/_logstress user/_forphan user/_d
orphan user/_sandbox user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap bl
ocks 1) blocks 1953 total 2000
ballocc: first 1009 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kern
el -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=fa
lse -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-b
lk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sandbox 32768 - cat README
cat: cannot open README
$ |

```

```

ramisk@Ramis-Laptop: ~/xv6
ballocc: first 1129 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kern
el -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=fa
lse -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-b
lk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ ./grade-lab-syscall sandbox_mask
exec ./grade-lab-syscall failed
$ sandbox 32768 - cat README
cat: cannot open README
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson'
s Unix
Version 6 (v6). xv6 loosely follows the structure and style of
v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (
Peer
(kaashoek,rtm@mit.edu). The main purpose of xv6 is as a teachin
g

```

```

== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (3.4s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.5s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.2s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.7s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.1s)
== Test attack ==
$ make qemu-gdb
attack: OK (0.9s)
== Test time ==
time: OK
Score: 45/45

```

```

ramisk@Ramis-Laptop: ~/xv6
-fno-stack-protector -fno-pie -no-pie -c -o user/secret.o use
r/secret.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o us
er/_secret user/secret.o user/ulib.o user/usys.o user/printf.o u
ser/umalloc.o
riscv64-linux-gnu-objdump -S user/_secret > user/secret.asm
riscv64-linux-gnu-objdump -t user/_secret | sed '1,/SYMBOL TABLE
/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_
forktest user/_grep user/_init user/_kill user/_ln user/_ls user
/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_g
rind user/_wc user/_zombie user/_logstress user/_forphan user/_d
orphan user/_sandbox user/_attack user/_attack_debug user/_attac
k_final user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap bl
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lse -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-b
lk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ secret xyzyz
$ attack
xyzyz
$ |

```